

Nothin' But Data

Probabilistic Forecasting of the 2026 NCAA Tournaments

Team Members Info

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Problem Description

The primary objective of this project is to build **a robust predictive system** capable of forecasting the **outcomes of every possible matchup** in men's **2026 NCAA college basketball tournaments**. These tournaments are famous for their single-elimination bracket format and high frequency of unpredictable upsets. We won't be so arrogant as to assume we can predict the outcome of each individual match, but our goal is to be precise enough in the probability forecasting that our algorithm will be the **safest bet in the long run**.

Our hypothesis is that by leveraging statistical models and machine learning techniques, we can identify underlying patterns hiding in historical performance, team capabilities and behavior or geographic and demographic context. The success of our system can be evaluated using the **actual game outcomes** as they unfolded during the 2026 tournaments.

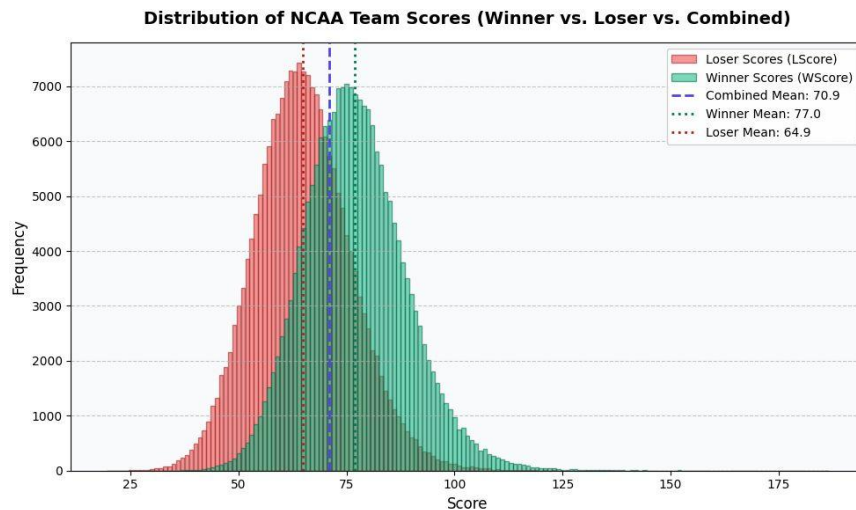
Data Description

Our dataset was obtained from Kaggle's official "[March Machine Learning Mania 2026](#)" competition. It consists of approximately 7 million total records (rows) spread across 21 separate files. The combined storage of the dataset is 157 MB. The files comprehensively capture historical Division I college basketball game data and contains:

- **Core Game Results:** Outcomes (winning/losing teams, final scores, and locations) for both regular-season and NCAA tournament games dating back to 1985.
- **Detailed Box Scores:** Granular, game-by-game statistical metrics (such as field goal attempts/makes, 3-pointers, offensive/defensive rebounds, assists, turnovers, steals, and blocks) available from 2003 onwards.
- **Public Rankings (MMasseyOrdinals):** The largest file in the dataset (123.15 MB, over 5.8 million records), containing historical weekly team ordinal rankings from dozens of established rating systems. This is the traditional point of view.
- **Supplementary Meta-data:** City geographic data, coaching timelines, conference affiliations, tournament bracket slotting structures, and secondary post-season tournament outcomes (e.g., NIT, WNIT).

Visualization #1 - Distribution of NCAA Team Scores

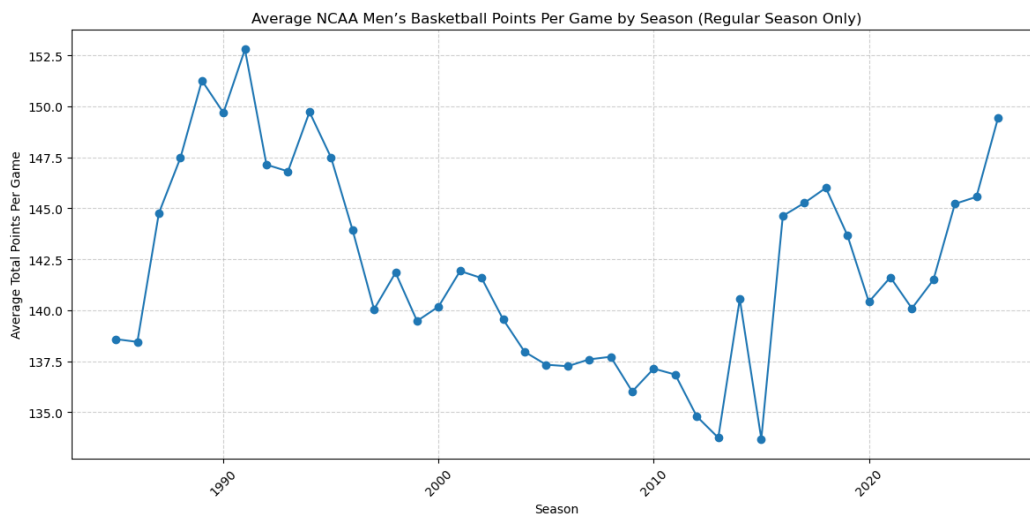
Expectation: We expected NCAA team scores to follow an approximately bell-shaped distribution centered around a realistic basketball scoring range of roughly 65–80 points. We also expected winning teams to score noticeably more points on average than losing teams.



Results: The visualization supports our expectation. Both winner and loser scores form roughly symmetric distributions, with winner scores centered around 77.0 points and loser scores centered around 64.9 points. The distributions overlap substantially, reflecting the fact that many games are competitive rather than complete blowouts. Extreme scores occur only rarely in the tails. Additionally, all observed scores fall within a realistic range for college basketball, providing a useful sanity check on the dataset.

Visualization #2 - Average Points Per Game by Season

Expectation: We expected average scoring to vary across decades rather than remain constant. Changes in offensive and defensive strategies, pace of play, coaching philosophies, and NCAA rule changes may influence the number of points scored in a typical game.



Results: The visualization confirms this expectation. Average points per game fluctuate noticeably across seasons, ranging from approximately 134 to 153 points. Scoring was relatively high during the late 1980s and early 1990s, declined through the 2000s and early 2010s, and increased again in recent years. This suggests that the NCAA scoring environment evolves over time and that season context may be important when comparing teams from different eras.